

QICHEN SONG

12 Oxford Street, Cambridge, MA 02138 · qichensong@g.harvard.edu · qichensong.github.io

EDUCATION AND EXPERIENCES

Harvard University 1/2022 — present

Harvard Quantum Initiative Postdoctoral Fellow (Advisor: Jarad A. Mason)

Research: Thermal transport in phase-change inorganic materials

Massachusetts Institute of Technology 9/2015 — 1/2022

Ph.D. in Mechanical Engineering (Advisor: Gang Chen)

Thesis: Phonon and electron transport through interfaces and disordered structures

S.M. in Mechanical Engineering (Advisor: Gang Chen)

Thesis: *Ab initio* study of electron transport in lead telluride

Huazhong University of Science and Technology 9/2011 — 6/2015

B.S. with Honors in Thermal Energy and Power Engineering (Advisor: Nuo Yang)

Thesis: On the effect of interlayer interaction on the thermal conductivity in folded graphene

RESEARCH INTERESTS

1. Understanding, manipulating, and optimizing the conversion, transfer, and storage of energy in stimuli-responsive and quantum materials.
2. Solid-state cooling systems using novel caloric materials.

Specializations: Transport physics, phase-change materials, laser spectroscopy, first-principles calculations

FUNDING AND AWARDS

Research Grant, Fourier Scientific LLC 2025 — 2026

Project: Non-contact measurement of thermal conductivity and heat capacity in single crystals via optical reflectance

Harvard Quantum Initiative Postdoctoral Prize 2022 — 2025

Kaufman Teaching Certificate Program 2020

MIT Quantum Hackathon Creative Award 2020

Warren M. Rohsenow Fellowship 2015 — 2016

National Scholarship (three times) 2012 & 2013 & 2014

PUBLICATIONS



- (34) **Qichen Song**, Rui Pan, Jungwoo Shin, Aaron J. Schmidt, Asegun Henry, Hong Lu, Gang Chen*, “Phonon Anderson localization in aperiodic superlattices up to 200 K”, *in preparation*
- (33) Rahil Ukani, **Qichen Song**, Alex D. Christodoulides, Caleb Stamper; Jinyoung Seo, Yukyung Moon, Dehong Yu, Catherine Thai, Ryan D. McGillicuddy, Jonathan A. Malen*, and Jarad A. Mason*, “The effect of order–disorder transitions on thermal transport in a two-dimensional hybrid perovskite”, to be submitted to *J. Am. Chem. Soc.*
- (32) Yukyung Moon, **Qichen Song**, Rahil Ukani, and Jarad A. Mason*, “Guest-controlled thermal transport in metal–organic frameworks”, to be submitted to *J. Am. Chem. Soc.*
- (31) Vidhya M. Dev, Kevin Miura, **Qichen Song**, Jason D. Braun, Faith E. Chen, Jinyoung Seo, Daniel W. Laorenza, and Jarad A. Mason*, “Tailoring spin-crossover transitions in Fe[HB(tz)₃]₂ via co-crystallization for ultra-low

temperature barocaloric cooling”, to be submitted to *J. Am. Chem. Soc.*, **2025**

- (30) Lucy S. Nathwani, Anne Ruperto, Ashvini Vallipuram, Abigail Y. Jiang, Grace A. Pan, Dan Ferenc Segedin, Ari B. Turkiewicz, Charles M. Brooks, Jarad A. Mason*, **Qichen Song***, and Julia A. Mundy*, “Observation of microscopic domain effects in the metal-insulator transition of thin-film NdNiO₃”, *Nano Lett.*, **2026**, 26, 4027–4033
- (29) **Qichen Song**, Rahil Ukani, Vidhya Dev, Jungwoo Shin, Jinyoung Seo, Hong Ki Kim, Caleb Stamper, Ryan McGillicuddy, Catherine Thai, Dehong Yu, Gang Chen, Jarad A. Mason*, “Reversible regulation of thermal conductivity through spin-crossover transitions”, accepted in *J. Am. Chem. Soc.*, **2026** [[Manuscript](#)] [[Supplementary Information](#)] [[Code & Data](#)]
- (28) **Qichen Song***, Sorren Warkander, Samuel C. Huberman*, “Probing carrier and phonon transport in semiconductors all at once through frequency-domain photorefectance”, *Phys. Rev. Applied*, **2024**, 21, 034044
- (27) Lenan Zhang*, Yang Zhong, Xiangyu Li, Ji-Hoon Park, **Qichen Song**, Long Li, Liang Guo, Jing Kong*, and Gang Chen*, “Effect of twist angle on interfacial thermal transport in two-dimensional bilayers”, *Nano Lett.*, **2023**, 17, 7790–7796
- (26) Ryotaro Okabe[†]*, Abhijatmedhi Chotrattanapituk[†], Artittaya Boonkird, Nina Andrejevic, Xiang Fu, Tommi S. Jaakkola, **Qichen Song**, Thanh Nguyen, Nathan Drucker, Sai Mu, Bolin Liao, Yongqiang Cheng, Mingda Li*, “Virtual node graph neural network for full phonon prediction”, *Nat. Comput. Sci.*, **2024**, 4, 522–531
- (25) Zhantao Chen*, Xiaozhe Shen*, Nina Andrejevic, Tongtong Liu, Duan Luo, Thanh Nguyen, Nathan C. Drucker, Michael E. Kozina, **Qichen Song**, Chengyun Hua, Gang Chen, Xijie Wang, Jing Kong, Mingda Li*, “Panoramic mapping of phonon transport from ultrafast electron diffraction and scientific machine learning”, *Adv. Mater.*, **2022**, 2206997
- (24) Yu Dai, Wenjiang Zhou, Hyun-Jung Kim, **Qichen Song**, Xin Qian, Te-Huan Liu*, Ronggui Yang*, “Simultaneous enhancement in electrical conductivity and Seebeck coefficient by single- to double-valley transition in a Dirac-like band”, *npj Comput. Mater.*, **2022**, 8, 234
- (23) Chaitanya A. Garde[†], Xingxu Yan[†], **Qichen Song**, Jie Li, Lei Gu, Toshihiro Aoki, Sheng-Wei Lee, Gang Chen, Ruiqian Wu, Xiaoqing Pan*, “Nanoscale imaging of phonon dynamics by electron microscopy”, *Nature*, **2022**, 606, 292–297
- (22) **Qichen Song** and Gang Chen*, “Significant reduction in semiconductor interface resistance via interfacial atomic mixing”, *Phys. Rev. B*, **2022**, 105, 195306
- (21) Jiawei Zhou*, Hangtian Zhu, **Qichen Song**, Zhiwei Ding, Jun Mao, Zhifeng Ren, Gang Chen*, “Mobility enhancement in heavily doped semiconductors via electron cloaking”, *Nat. Commun.*, **2022**, 13, 2482
- (20) Lenan Zhang, Yong Zhong, Xin Qian, **Qichen Song**, Jiawei Zhou, Long Li, Liang Guo*, Gang Chen*, and Evelyn N. Wang*, “Toward optimal heat transfer of 2D–3D heterostructures via van der Waals binding effects”, *ACS Appl. Mater. Interfaces*, **2021**, 13, 38
- (19) **Qichen Song*** and Gang Chen*, “Evaluation of diffuse mismatch model for phonon scattering at disordered interfaces”, *Phys. Rev. B*, **2021**, 104, 085310
- (18) Chaitanya Gadre, Xingxu Yan, **Qichen Song**, Gang Chen, Xiaoqing Pan, “Phonon reflections from nanostructured interfaces imaged by momentum-averaged and resolved vibrational EELS”, *Microsc. Microanal.*, **2021**, 27 (S1), 1204–1206
- (17) Thanh Nguyen, Nina Andrejevic, Hoi Chun Po, **Qichen Song**, Yoichiro Tsurimaki, Nathan C. Drucker, Ahmet Alatas, Esen E. Alp, Bogdan M. Leu, Alessandro Cunsolo, Yong Q. Cai, Lijun Wu, Joseph A. Garlow, Yimei Zhu,

- Hong Lu, Arthur C. Gossar Alexander A. Poretzky, David B. Geohegan, Shengxi Huang*, Mingda Li*, “Signature of many-body localization of phonons in strongly disordered superlattices”, *Nano Lett.*, **2021**, 17, 7419–7425
- (16) Haozhe Wang, Zhenpeng Yao, Gang Seob Jung, **Qichen Song**, Marek Hempel, Tomas Palacios, Gang Chen, Markus J. Buehler, Alan Aspuru-Guzik, Jing Kong*, “Frank-van der Merwe growth in bilayer graphene”, *Matter*, **2021**, 4, 10, 3339–3353
- (15) Wuyang Ren[†], **Qichen Song**[†], Hangtian Zhu, Jun Mao, Li You, Geethal A. Gamage, Jiawei Zhou, Ting Zhou, Jing Jiang, Chao Wang, Jun Luo, Jiang Wu, Zhiming Wang*, Gang Chen*, Zhifeng Ren*, “Intermediate-level doping strategy to simultaneously optimize power factor and phonon thermal conductivity for improving thermoelectric figure of merit”, *Mater. Today Phys.*, **2020**, 15, 100250
- (14) Qiyang Lu, Samuel Huberman, Hantao Zhang, **Qichen Song**, Jiayue Wang, Gulin Vardar, Adrian Hunt, Iradwikanari Waluyo, Gang Chen* and Bilge Yildiz*, “Bi-directional tuning of thermal transport in SrCoO_x with electrochemically induced phase transitions”, *Nat. Mater.*, **2020**, 19, 655–662
- (13) Ke Chen[†], Bai Song[†]*, Navaneetha K. Ravichandran, Qiye Zheng, Xi Chen, Hwijong Lee, Haoran Sun, Sheng Li, Geethal A. Gamage, Fei Tian, Zhiwei Ding, **Qichen Song**, Akash Rai, Hanlin Wu, Pawan Koirala, Aaron J. Schmidt, Kenji Watanabe, Bing Lv, Zhifeng Ren, Li Shi, David G. Cahill, Takashi Taniguchi, David Broido* and Gang Chen*, “Ultrahigh thermal conductivity in isotope-enriched cubic boron nitride”, *Science*, **2020**, 367, 6477
- (12) Chang Liu, **Qichen Song**, Jianan Chen, Xinhao Li, Jingxuan Cai, Zhouguang Lu, Wendi Li, Nicholas X. Fang*, Shien-Ping Feng*, “Electromagnetic and chemical enhancements of surface-enhanced Raman scattering spectra from Cu₂O hexagonal nanoplates”, *Adv. Mater. Interfaces*, **2019**, 6, 17, 1900534
- (11) Te-Huan Liu, Bai Song, Laureen Meroueh, Zhiwei Ding, **Qichen Song**, Jiawei Zhou, Mingda Li, Gang Chen*, “Simultaneously high electron and hole mobilities in cubic boron-V compounds: BP, BAs and BSb”, *Phys. Rev. B: Rapid Communications*, **2018**, 98, 081203
- (10) Hangtian Zhu, Jun Mao, Yuwei Li, Jifeng Sun, Yumei Wang, Qing Zhu, Guannan Li, **Qichen Song**, Jiawei Zhou, Yuhao Fu, Ran He, T. Tong, Zihang Liu, Wuyang Ren, Li You, Zhiming Wang, Jun Luo, Andrei Sotnikov, Jiming Bao, Kornelius Nielsch, Gang Chen, David J. Singh* and Zhifeng Ren*, “Discovery of TaFeSb-based half-Heuslers with high thermoelectric performance”, *Nat. Commun.*, **2019**, 10, 270
- (9) Qian Zhang*, **Qichen Song**, Xinyu Wang, Jingying Sun, Qing Zhu, Keshab Dahal, Xi Lin, Feng Cao, Jiawei Zhou, Shuo Chen, Gang Chen, Jun Mao*, Zhifeng Ren*, “Deep defect level engineering: a strategy of optimizing the carrier concentration for high thermoelectric performance”, *Energy & Environmental Science*, **2018**, 11 (4), 933–940
- (8) Jiawei Zhou, Hangtian Zhu, Te-Huan Liu, **Qichen Song**, Ran He, Jun Mao, Zihang Liu, Wuyang Ren, Bolin Liao, David J. Singh, Zhifeng Ren*, Gang Chen*, “Large thermoelectric power factor from crystal symmetry-protected non-bonding orbital in half-Heuslers”, *Nat. Commun.*, **2018**, 9, 1721
- (7) Te-Huan Liu, Jiawei Zhou, Mingda Li, Zhiwei Ding, **Qichen Song**, Bolin Liao, Liang Fu, Gang Chen*, “Electron mean-free-path filtering in Dirac material for improved thermoelectric performance”, *Proc. Natl. Acad. Sci.*, **2018**, 115 (5), 879–884
- (6) Mingda Li[†]*, **Qichen Song**[†], Weiwei Zhao, Joseph A. Garlow, Te-Huan Liu, Lijun Wu, Yimei Zhu, Jagadeesh S. Moodera, Moses H. W. Chan, Gang Chen*, and Cui-Zu Chang*, “Dirac-electron-mediated magnetic proximity effect in topological insulator/magnetic insulator heterostructures”, *Phys. Rev. B: Rapid Communications*, **2017**, 96, 201301

- (5) **Qichen Song**, Te-Huan Liu, Jiawei Zhou, Zhiwei Ding, Gang Chen*, “*Ab initio* study of electron mean free paths and thermoelectric properties of lead telluride”, *Mater. Today Phys.*, **2017**, 2, 69–77
- (4) Meng An[†], **Qichen Song**[†], Xiaoxiang Yu, Zelin Jin, Dengke Ma, Baoling Huang, Nuo Yang*, “Generalized two-temperature model for coupled phonons”, *Nano Lett.*, **2017**, 17 (9), 5805–5810
- (3) Mingda Li*, **Qichen Song**, Te-Huan Liu, Laureen Meroueh, Gerald D. Mahan, Mildred. S. Dresselhaus, Gang Chen*, “Tailoring superconductivity with quantum dislocations”, *Nano Lett.*, **2017**, 17 (8), 4604–4610
- (2) **Qichen Song**, Jiawei Zhou, Laureen Meroueh, David Broido, Zhifeng Ren, Gang Chen*, “The effect of shallow vs. deep level doping on the performance of thermoelectric materials”, *Appl. Phys. Lett.*, **2016**, 109, 263902
- (1) **Qichen Song**[†], Meng An[†], Xiandong Chen, Zhan Peng, Jianfeng Zang, Nuo Yang*, “The adjustable thermal resistor by reversibly folding a graphene sheet”, *Nanoscale*, **2016**, 8, 14943–14949

PRESENTATIONS

- (9) Probing charge and heat transport simultaneously in thin films and single crystals via photo-reflectance, **2026** International Conference on Thermoelectrics, Houston, Texas
- (8) Reversible regulation of thermal conductivity through spin-crossover transitions, **2026** Emerging Scholars Forum, Singapore
- (7) Reversible regulation of thermal conductivity with spin state transitions, **2024** MRS Fall Meeting, Boston, Massachusetts
- (6) Bridging thermal engineering and chemistry with molecular materials, Meet the New Faculty Candidates Poster Session, **2024** MRS Fall Meeting, Boston, Massachusetts
- (5) Study of carrier and thermal transport in semiconductors using frequency domain thermoreflectance, **2023** MRS Spring Meeting, San Francisco, California
- (4) Manipulating electron and phonon flow across interfaces using structural randomness, **2023** APS March Meeting, Las Vegas, Nevada
- (3) Probing local heating and cooling at interfaces: a non-equilibrium Green’s function study, **2018** APS March Meeting, Los Angeles, California
- (2) *Ab initio* study of the effect of the grain boundary on electron transport in thermoelectric materials (poster), **2017** MRS Fall Meeting, Boston, Massachusetts
- (1) *Ab initio* study of electron transport in lead telluride, **2017** APS March meeting, New Orleans, Louisiana

SERVICES

Session Chair, Session 4. Machine learning-II, **2026**, the 42nd International Conference on Thermoelectrics

Session Chair, SF03.19 Emerging Thermal Materials–From Nanoscale Heat Transport, Devices and Applications, to Theories, **2023** MRS Fall Meeting

Journal Reviewer for *PRL*, *PRB*, *PR Applied*, *Sci. Adv.*, *Nano Lett.*, *Adv. Mater.*, *Adv. Funct. Mater.*, *Joule*, *Mater. Today Phys.*, *ACS Appl. Mater. Interfaces*, *Appl. Phys. Lett.*, *J. Phys. Chem. Lett.*, *J. Mater. Chem. C.*, *npj Comput. Mater.*, *IEEE T-DMR*, *IEEE T-TST*, *Proc. R. Soc. A.*, *J. Low Temp. Phys.*

Guest Editor, Special Issue on Atomic Layer Materials and Processes, *Micromachines*

Project Teach with Brighton High School and Dr. Martin Luther King, Jr. School

TEACHING

TA for 2.57 Nano-to-Macro Transport Processes, Fall **2019** (overall rating: 6.8 out of 7)

Instructor for Introduction to Pump-Probe Spectroscopy for Study of Thermal Transport in Materials, Independent

MENTORING

Rahil Ukani, Ph.D. candidate (G7) in the Department of Chemistry & Chemical Biology, Harvard University

Diffuson-like heat transport in dynamically disordered chains

Yukyung Moon, Ph.D. candidate (G5) in the Department of Chemistry & Chemical Biology, Harvard University

Bidirectional adjustment of thermal conductivity in metal-organic frameworks via gas-host interactions

Lucy Nathwani, undergraduate researcher (U4) in the Department of Physics, Harvard University

Tuning thermal transport in NdNiO₃ thin films with metal-insulator transitions

Ashvini Vallipuram, Ph.D. student (G2) in the Department of Physics, Harvard University

Engineered Thermal transport in V₂O₃ thin films

COURSES

MechE (major)

Advanced Fluid Mechanics, General Thermodynamics, Advanced Heat & Mass Transfer, Nano-to-Macro Transport Processes

Physics (minor)

Theory of Solids II, Relativistic Quantum Field Theory I, Relativistic Quantum Field Theory II, Statistical Mechanics I, Statistical Mechanics II

EECS

Applied Quantum & Statistical Physics, Physics for Solid-State Applications, Principles & Applications of Quantum Optics

MSE

Atomistic Computer Modeling of Materials

Math

Mathematical Methods in Nanophotonics, Computational Science & Engineering I

COMPUTER SKILLS

FORTRAN 90, Python, Qiskit, MATLAB, L^AT_EX, C++, LabVIEW

REFERENCES

Gang Chen

Carl Richard Soderberg Professor of Power Engineering of the Department of Mechanical Engineering
gchen2@mit.edu

Jarad A. Mason

Professor of Chemistry and Chemical Biology
mason@chemistry.harvard.edu

Mingda Li

Associate Professor of Nuclear Science and Engineering
mingda@mit.edu

Julia A. Mundy

John L. Loeb Associate Professor of the Natural Sciences and of Engineering and Applied Sciences
mundy@fas.harvard.edu